Warm-up

(1) Find $\displaystyle\int \frac{1}{2+x^2}\,dx$.

(2) Is $\displaystyle\int_1^3 \frac{1}{x}\,dx$ greater than, equal to, or less than 1?

Medium

(1) Compute $\arcsin\left(-\sqrt{3}/2\right)$.

(2) Compute $\displaystyle\frac{d}{dx}\frac{1}{\arcsin(x^2)}$.

Evaluate the following integrals:

$$(3)\ \int_0^2 \frac{e^{2x}}{e^{4x}+1}\,dx, \quad (4)\ \int \cot(5x)\,dx, \quad \text{and} \quad (5)\ \int \frac{1}{x\sqrt{4-\ln(x)^2}}\,dx.$$

Stretch

(1) If we were to invert $\sec(x)$ to define $\operatorname{arcsec}(x)$, what interval should we first restrict it to? (There are multiple answers here, but only one is *best*.) Compute $\frac{d}{dx}\operatorname{arcsec}(x)$.

(2) Use the figure below to compute $\arctan(1/2) + \arctan(1/3)$.